Fares Nagy Eleissawy *Senior GIS Developer*

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SUMMARY

Senior GIS Developer with 6+ years of experience designing, building, and owning enterprise GIS applications, web GIS solutions, and spatial databases on the Esri ArcGIS platform (ArcGIS Enterprise, Portal, ArcGIS Online) and open-source stacks. Built national-scale flood simulation and dam-break analysis platforms for Saudi government agencies and founded OnePin, a global building identification system. Expert in ArcGIS Maps SDK for JavaScript, React/TypeScript, Python/ArcPy, REST API integration, and GIS service publishing, with full ownership of production support and performance optimization. Delivered large-scale solutions for government and enterprise clients across water, environment, telecommunications, and urban planning sectors.

PROFESSIONAL EXPERIENCE

Senior GIS Developer | Evara Group (Consultant) | Full-time | Riyadh, KSA | Dec 2025 - Present

- Designed, developed, and engineered a **national-scale flood simulation and hazard assessment platform** for Saudi Arabia's **Ministry of Environment, Water and Agriculture (MEWA)**, delivering a core digital capability that directly advances **Saudi Vision 2030** objectives for water security and disaster resilience.
- Architected the platform end to end with **ArcGIS JavaScript API** 4.x, **React**, **Node.js**, and TypeScript, building **real-time** dam-break scenario visualization and **3D terrain modeling** for emergency response planning.
- Led **solution architecture**, technical design, and **effort estimation**, and drove **code reviews** within an **Agile/Scrum** team of business analysts, UI/UX, and QA.
- Built a hazard assessment **dashboard** that fuses **satellite imagery**, topographic data, and real-time sensor networks to predict flood impacts on infrastructure.
- Engineered a building damage assessment module that automatically calculates structural impacts from flood parameters, generating **actionable intelligence** for disaster preparedness.
- Created **time-based simulation** controls that let decision-makers analyze flood progression at any interval for evidence-based resource allocation.

GIS Developer & Technical Support Engineer | Full-time | Syadtech | KSA | Jan 2024 - Dec 2025

- Developed the **Madinati** mobile GIS app for the **Eastern Province Municipality (Amanah)** on **ESRI API JavaScript 3.4**, delivering the upgrade within a 2-month timeline for a public app available on iOS and Android with **50,000+ downloads** on Google Play (play.google.com/store/apps/details?id=com.eamana.webgis).
- Developed and maintained web GIS applications on the **Eastern Province Municipality GIS Geoportal**, serving public and municipal map services (eamana.balady.gov.sa/gis-home/Apps).
- Cut system downtime by monitoring and troubleshooting Enterprise **ArcGIS** applications built with **React**, resolving 95% of issues within SLA.
- Owned **production support** and **root cause analysis** for GIS applications, map services, and spatial databases, and managed deployments across Development, Testing, UAT, and Production.
- Accelerated development teams' integration work by authoring **API/SDK documentation** and leading hands-on technical guidance.

GIS Developer | Etisalat Egypt | Full-time | Egypt | Jul 2023 - Dec 2023

- Optimized **.NET** and **ArcGIS Enterprise** GIS platforms and APIs, accelerating query performance through database indexing and **spatial query** tuning.
- Maintained mission-critical location-based services at near 100% uptime as the **technical owner** for their support.
- Integrated **GIS functionality into telecommunications applications**, guiding developer teams on spatial analysis and geocoding services.

GIS Specialist | Global Appraisal Tech | Full-time | Egypt | Jul 2019 - Jul 2022

- Developed enterprise GIS applications with **ESRI Experience Builder**, **Web AppBuilder**, and **Dashboards**, serving 100+ concurrent users.
- Reduced average ticket resolution time by running **software support** for the GIS applications.
- Authored **API documentation**, **technical guides**, and **code samples** that streamlined developer onboarding.

GIS Developer - R&D | Itechs-DI | Full-time / Part-time | Egypt | Dec 2023 - Dec 2025

- Designed **off-road routing** algorithms that improved route calculation accuracy for military operations.
- Architected a NATO-compliant **real-time geospatial** application with **Django**, **Angular 15/17**, **ArcGIS API**, and **WebSocket** communication for offline tactical scenarios.
- Optimized **geospatial data processing pipelines** and resolved API issues for internal teams, reducing processing time.

Teaching Assistant - GIS Department | Information Technology Institute | Part-time | Egypt | Jul 2022 - Present

- Teach spatial data analysis and GIS development to 200+ students, and co-develop curriculum for **ArcGIS** and **open-source GIS** training programs.

GIS Developer/Consultant | Part-time | Multiple Firms

ESense Software - 3 months

- Optimized and debugged a mobile GIS application built with **Xamarin** that consumes **ArcGIS** services through **.NET Web APIs**, and evaluated its requirements.

Infasme - 3 months

- Optimized and debugged **KFH - Kuwait Finance House**'s **full-stack** stock market application built with **.NET** and **Angular**.
- Built an **automated Esri Dashboards creation** system on ArcGIS Enterprise using **Python** and AI models.

Amanat Al-Madinah Al-Monawarah - 3 months

- Engineered a real-time **ETL** pipeline (**Python Sync Engine**) between an enterprise **Oracle Database** and **ArcGIS Enterprise Feature Services**, keeping feature services **type-safe**, **schema-safe**, and **real-time**.

EDUCATION

Bachelor of Science in Geophysics

Tanta University | Tanta, Egypt | 2015 - 2019

Post-Graduate Diploma in GIS Development

Professional Training Program, GIS Track - Information Technology Institute (ITI) | 2022 - 2023

SKILLS

Programming & Development: JavaScript, TypeScript, HTML5, CSS, Python, ArcPy, C#/.NET, Java, SQL, Django, Angular 15/17, React, Next.js, ASP.NET MVC, Node.js

Esri ArcGIS Platform: ArcGIS Maps SDK for JavaScript (3.x/4.x), ArcGIS Enterprise, ArcGIS Server, ArcGIS Portal, ArcGIS Online, ArcGIS Pro, ArcGIS Pro SDK, Experience Builder, Web AppBuilder, ArcGIS Dashboards, ArcGIS Field Maps

GIS Services & Administration: Publishing and managing map, feature, and geoprocessing services; ArcGIS REST API; GIS security, authentication and authorization (users, roles, groups, permissions); deployment across Dev, Test, UAT, and Production

GIS Concepts & Analysis: Web GIS development, interactive web mapping applications, enterprise GIS, spatial analysis, geoprocessing, coordinate systems and projections, geocoding, spatial data modeling, geodatabases, 3D analysis, hydrological modeling

Open Standards & Interoperability: GeoServer, QGIS, OpenLayers, OGC standards (WMS, WFS, WMTS, WCS), GeoJSON; interoperability between Esri and open-source GIS platforms

Databases & APIs: Enterprise systems integration (REST APIs, ArcGIS REST Services, database integrations), PostgreSQL/PostGIS, MS SQL Server, Oracle Spatial, RESTful API development, WebSocket, PWA, HERE Location Services, Geolocation APIs

DevOps, Cloud & Tools: Git, Jira, Docker

Methodologies: Agile/Scrum, SDLC, code reviews, technical documentation, mentoring

Remote Sensing & Analysis: Deep Learning for Imagery Analysis, Land Surface Temperature Analysis, DEM Processing, Image Processing

PROJECTS

OnePin - Global Building Identification Mobile Application

Developed innovative mobile application that assigns unique 16-digit identification codes to 3D buildings globally. Utilizes advanced geospatial algorithms analyzing building footprints, 3D geometry, and spatial relationships to generate permanent, universal building identifiers. Platform enables seamless building identification across borders supporting emergency services, logistics, property management, and urban planning worldwide.

GeoIntel - AI Platform for GIS Services

Live: geointel-theta.vercel.app

AI-powered web platform (**Next.js**, **ArcGIS Maps SDK**, **Claude API**) for exploring **ArcGIS** and **GeoServer** services through natural-language questions. It includes spatial and impact analysis, interactive map-linked dashboards in the style of ArcGIS Dashboards, and AI-written reports. Deployed on **Vercel** with **Neon Postgres**, email sign-in, and per-user usage metering.

Aircraft Global Explorer - 3D Flight Simulator & World Explorer (Web / PWA)

Live: faresnagyeleissawy.github.io/aircraft-global-explorer/ | Code: github.com/faresnagyeleissawy/aircraft-global-explorer

Browser-based 3D flight simulator for flying a plane, helicopter, or fighter jet over real cities worldwide, with real airports, runways, 3D buildings, and satellite imagery.

- Built a 3D globe in **CesiumJS** with worldwide OSM 3D buildings (custom **GLSL** shader), terrain, and sun shadows; implemented 6-DOF flight physics and an airport-to-airport autopilot over 3,000+ real airports (**OurAirports**) with great-circle routing.
- Integrated search (**Nominatim / Photon**), road routing (**OSRM**), and live **ADS-B** air traffic; shipped as an installable **PWA** in plain JavaScript (ES modules) with Node unit tests, deployed on **GitHub Pages**.

SDG-Aligned Renewable Energy Dashboard

ESRI Web AppBuilder dashboard tracking renewable energy indicators, with secure ASP.NET Web APIs and identity authentication.

CERTIFICATIONS & AWARDS

Professional Certificate in GIS Development - Information Technology Institute (ITI) | 2023

Winner - Shell N-Xplorers Program | Multiple second-place finishes in national and international GIS competitions | 2020